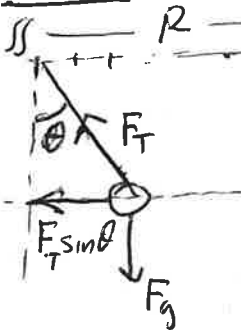
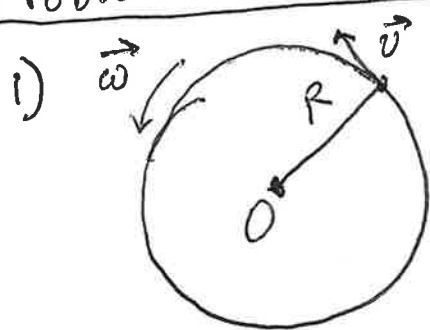


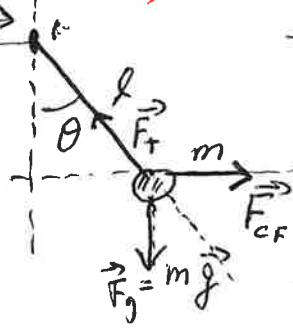
Problem Set #3

PART A (SOLUTION S)



Non-inertial frame

at equilibrium
 $\rightarrow \frac{d^2 \vec{r}}{dt^2} = 0$
 body



$$m \frac{d^2 \vec{r}}{dt^2} \Big|_{\text{body}} = \vec{F}_g + \vec{F}_T - m \frac{d^2 \vec{R}}{dt^2} \Big|_{\text{inertial}}$$

$\downarrow$ (pendulum at equilibrium) $\downarrow$ (horizontal component of tension) $\downarrow$ centripetal force

$$- m \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

x-component

$$\Rightarrow 0 = F_T \sin \theta - \frac{m v^2}{R} - m \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

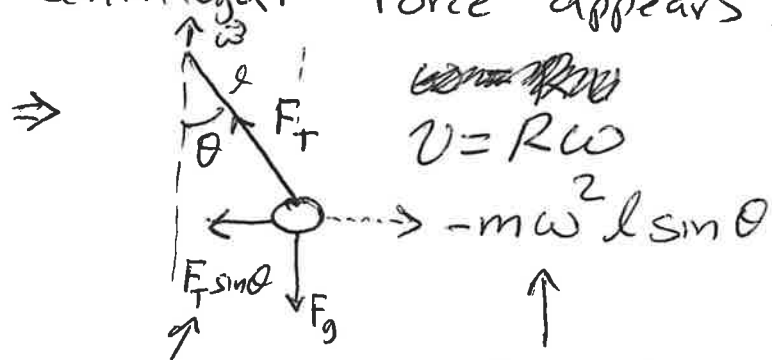
inertial

PART B

Inertial Frame: Forces present are $\vec{F}_T$ and $\vec{F}_g$

Non-Inertial Frame: Forces present are $\vec{F}_T$, $\vec{F}_g$ and $\vec{F}_{cf}$

However, the horizontal component of tension precisely cancels out the $-m \frac{d^2 \vec{R}}{dt^2} \Big|_{\text{inertial}}$ term, hence, there is NO centripetal force as seen by non-inertial observer, but rather a centrifugal force appears.



Shows up
in inertial frame
only

Shows up
in non-inertial
frame only

$$F_g = F_T \cos \theta$$

$$\Rightarrow \boxed{mg = F_T \cos \theta}$$

$$\Rightarrow \boxed{F_T \sin \theta = m \omega^2 l \sin \theta}$$

$$\Rightarrow \boxed{F_T = m \omega^2 l}$$

$$= m \frac{v^2}{R^2} l$$

1) PART C

To determine the equilibrium angle θ

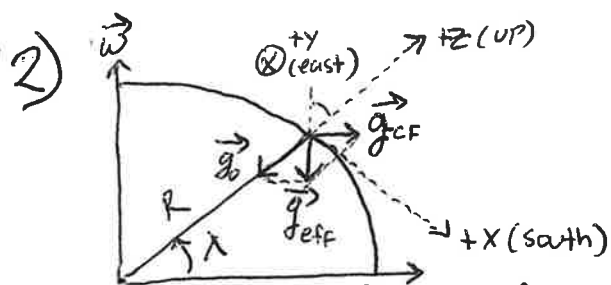
$$\rightarrow F_T \cos \theta = mg \rightarrow \cos \theta = \frac{mg}{F_T} = \frac{mg}{m\omega^2 l}$$

$$\Rightarrow \boxed{\theta = \cos^{-1} \left[\frac{g}{\omega^2 l} \right]} \Rightarrow \text{or}$$

or

$$F_T \sin \theta = \frac{mv^2}{R} \rightarrow \sin \theta = \frac{mv^2}{F_T R} = \frac{mv^2 l^2}{mv^2 l R}$$

$$\rightarrow \boxed{\theta = \sin^{-1} \left[\frac{R}{l} \right]}$$



$$\vec{g}_{eff} = \vec{g}_0 + \vec{g}_{cf}$$

PART A

$$\vec{g}_0 = -g_0 \hat{z}, \quad \vec{g}_{cf} = -\vec{\omega} \times (\vec{\omega} \times \vec{R})$$

$$\omega R \sin\left(\frac{\pi}{2} - \lambda\right) \hat{y} \cos \lambda$$

Let $\vec{\omega} = \omega_x \hat{x} + \omega_y \hat{y} + \omega_z \hat{z}$

$$-\vec{\omega} \times (\omega R \cos \lambda) \hat{y} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -\omega_x & -\omega_y & -\omega_z \\ 0 & \omega R \cos \lambda & 0 \end{vmatrix}$$

$$\vec{g}_{cf} = \hat{x} (\omega_z \omega R \cos \lambda) + \hat{z} (-\omega_x \omega R \cos \lambda)$$

$$\vec{g}_{eff} = \omega^2 R \sin \lambda \cos \lambda \hat{x} + (\omega^2 R \cos^2 \lambda - g_0) \hat{z}$$

$$|\vec{g}_{eff}| = (\vec{g}_{eff} \cdot \vec{g}_{eff})^{1/2}$$

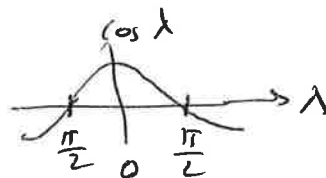
$$= (\omega^4 R^2 \sin^2 \lambda \cos^2 \lambda + \omega^4 R^2 \cos^4 \lambda - 2 \omega^2 R g_0 \cos^2 \lambda + g_0^2)^{1/2}$$

$$= [\omega^4 R^2 \cos^2 \lambda (\sin^2 \lambda + \cos^2 \lambda) - 2 \omega^2 R g_0 \cos^2 \lambda + g_0^2]^{1/2}$$

$$g_{eff} = [\cos^2 \lambda (\omega^4 R^2 - 2 \omega^2 R g_0) + g_0^2]^{1/2} \equiv g_{eff}(\lambda)$$

PART B

$$g_{eff}(\lambda = \pm 90^\circ) = \sqrt{g_0^2} = g_0 \Rightarrow \cos^2 90^\circ = 0$$



$$g_{eff}(\lambda = 0) = (\omega^4 R^2 - 2 \omega^2 R g_0 + g_0^2)^{1/2}$$

$$= [(g_0 - \omega^2 R)^2]^{1/2}$$

$$\Rightarrow g_{eff}(\lambda = 0) = g_0 - \omega^2 R$$

~~PART~~ 2) PART C

$$\underbrace{g_{\text{eff}}^{(\text{Poles})} - g_{\text{eff}}^{(\text{equator})}}_{\downarrow} = g_0 - g_0 + \omega^2 R$$

$$= +\omega^2 R, \quad \omega_e \approx 7.29 \times 10^{-5} \text{ rad/s}$$

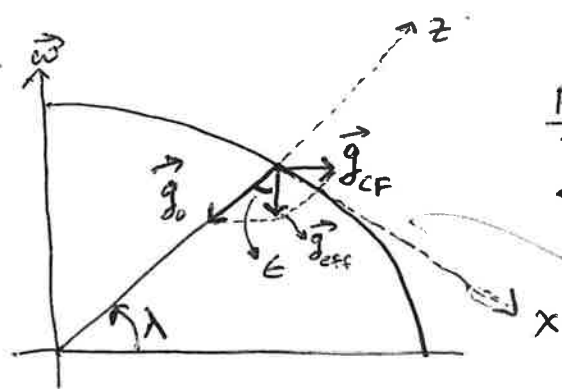
$$R_e \approx 6.38 \times 10^6 \text{ m}$$

$$|\Delta g_{\text{eff}}| = (7.29 \times 10^{-5})^2 \cdot 6.38 \times 10^6$$

$$\boxed{\Delta g_{\text{eff}} = 0.0339 \text{ m/s}^2}$$

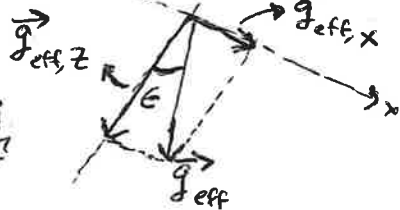
$$\frac{\Delta g_{\text{eff}}}{g_0} = 3.45 \times 10^{-3} \quad \text{or } \underline{\underline{\sim 0.346\%}}$$

3)



Recall:

$$\vec{g}_{\text{eff}} = \omega_e^2 R_e \sin \lambda \cos \lambda \hat{x} + (\omega_e^2 R_e \cos^2 \lambda - g_0) \hat{z}$$

ZOOM-IN of $\vec{g}_{\text{eff}}$ vector components

$$\tan \epsilon = \frac{|\vec{g}_{\text{eff},x}|}{|\vec{g}_{\text{eff},z}|} = \frac{\omega_e^2 R_e \sin \lambda \cos \lambda}{|\omega_e^2 R_e \cos^2 \lambda - g_0|}$$

$$\downarrow$$

$$\epsilon \ll 1$$

$$\tan \epsilon \approx \epsilon$$

$$\rightarrow |\omega_e^2 R_e \cos^2 \lambda - g_0| = \begin{cases} \omega_e^2 R_e \cos^2 \lambda - g_0, & \text{if } \omega_e^2 R_e \cos^2 \lambda > g_0 \\ -(\omega_e^2 R_e \cos^2 \lambda - g_0), & \text{if } g_0 > \omega_e^2 R_e \cos^2 \lambda \end{cases}$$

Since $\omega \approx \omega_e \ll 1$

$$\Rightarrow g_0 > \omega_e^2 R_e \cos^2 \lambda$$

(take the "+" side of the absolute value)

$$\Rightarrow \epsilon \approx \frac{\omega_e^2 R_e \sin \lambda \cos \lambda}{g_0 - \omega_e^2 R_e \cos^2 \lambda}$$

Determine the numerical value of ϵ_{max} :

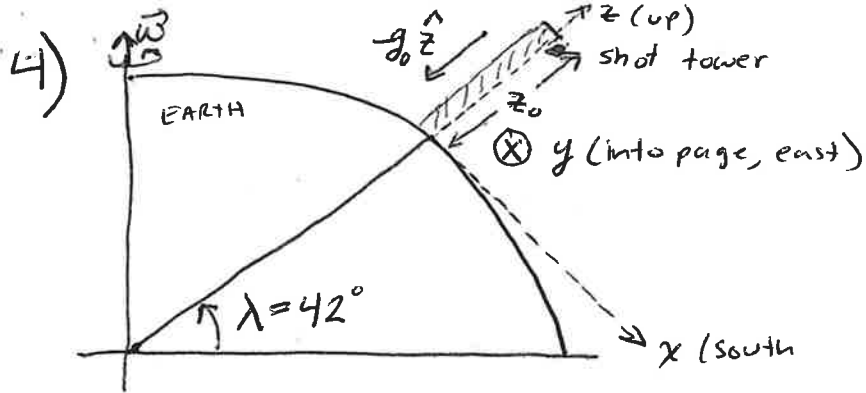
$$\epsilon_{\text{max}} \rightarrow \sin \lambda = \cos \lambda \rightarrow \lambda = \frac{\pi}{4} \text{ (or } 45^\circ)$$

$$\Rightarrow \text{for } \omega_e = 7.29 \times 10^{-5} \text{ s}^{-1}, R_e = 6.38 \times 10^6 \text{ m}, g_0 = 9.8 \text{ m/s}^2$$

$$\Rightarrow \epsilon_{\text{max}} \approx 1.73 \times 10^{-3} \text{ rad}, \quad 1 \text{ arcsec} \approx 4.8 \times 10^{-6} \text{ rad.}$$

$$\approx 1.73 \times 10^{-3} \text{ rad} \times \frac{1 \text{ arcsec}}{4.8 \times 10^{-6} \text{ rad}}$$

$$\Rightarrow \epsilon_{\text{max}} \approx 361 \text{ arcsec}$$



$$\vec{a}_{\text{tot}} = \vec{g}_0 + \vec{a}_{\text{cor}}$$

$$\vec{g}_0 = -g_0 \hat{z}$$

$$\vec{a}_{\text{cor}} = -2\vec{\omega} \times \vec{v}$$

Recall (from problem 2): $\vec{\omega} = -\omega \cos \lambda \hat{x} + \omega \sin \lambda \hat{z}$

generic $\vec{v}$ velocity of projectile $\Rightarrow \vec{v} = v_x \hat{x} + v_y \hat{y} + v_z \hat{z}$

Assuming "small deflections" from vertical (z) due to Coriolis term $\Rightarrow v_x \approx 0, v_y \approx 0 \Rightarrow$

NOTE: Only make this assumption when calculating Coriolis term

$$\Rightarrow \vec{\omega} \times \vec{v} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -\omega \cos \lambda & 0 & \omega \sin \lambda \\ 0 & 0 & v_z \end{vmatrix} = -\hat{y} (-\omega \cos \lambda \cdot v_z) = \omega \cos \lambda v_z \hat{y}$$

$$\Rightarrow \vec{a}_{\text{cor}} \approx a_{\text{cor}} \hat{y} = 2\omega \cos \lambda \cdot v_z \hat{y}$$

$$\Rightarrow \underline{a_{\text{cor}, y} = 2\omega \cos \lambda \cdot g_0 t}$$

determine v_z :

$$-g_0 = dv_z/dt$$

$$\Rightarrow \int_0^{v_z} dv_z = \int_0^t -g_0 dt$$

$$\underline{v_z(t) = -g_0 t}$$

Coriolis acceleration points east (+y), so the bullet is deflected eastward

$$a_{\text{cor}, y} = \frac{dv_y}{dt} = 2\omega \cos \lambda g_0 t$$

$$\Rightarrow \int_0^{v_y} dv_y = 2\omega \cos \lambda g_0 \int_0^t t dt$$

$$\Rightarrow \underline{v_y(t) = \omega \cos \lambda g_0 t^2}$$

4) Continued... Now we have the velocity of the projectile and will be able to determine deflection

$$\Rightarrow \underline{v_y(t) = \omega g_0 \cos \lambda t^2}$$

$$\underline{v_z(t) = -g_0 t}$$

$$v_y = \frac{dy}{dt} = \omega g_0 \cos \lambda t^2$$

$$v_z = \frac{dz}{dt} = -g_0 t$$

$$\int_0^y dy = \omega g_0 \cos \lambda \int_0^t t^2 dt$$

$$\int_{z_0}^z dz = -g_0 \int t dt$$

$$\boxed{y(t) = \frac{1}{3} \omega g_0 \cos \lambda t^3}$$

$$\boxed{z(t) = z_0 - \frac{1}{2} g_0 t^2}$$

Calculate projectile fall time, t_{FALL} :

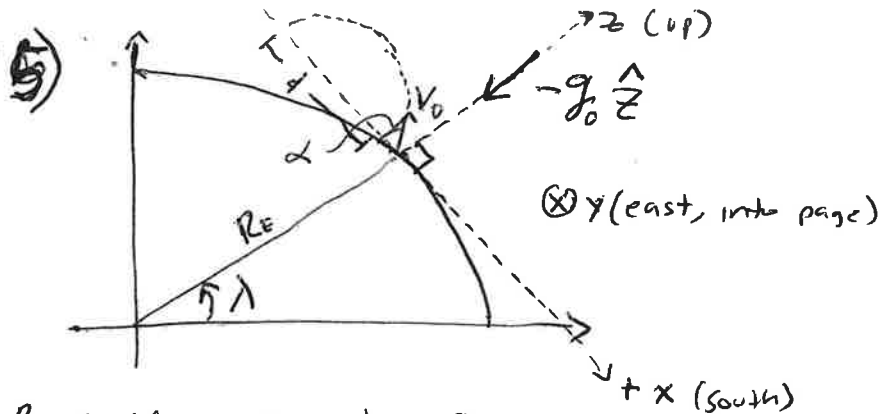
$$z(t_{\text{FALL}}) = z_0 - \frac{1}{2} g_0 t_{\text{FALL}}^2 = 0 \Rightarrow \boxed{t_{\text{FALL}} = \sqrt{\frac{2 z_0}{g_0}}}$$

Calculate projectile deflection eastward:

$$\boxed{y(t_{\text{FALL}}) = \frac{1}{3} \omega g_0 \cos \lambda \cdot \left(\frac{2 z_0}{g_0} \right)^{3/2}}$$

For $\omega = \omega_e = 7.3 \times 10^{-5} \text{ rad/s}$ $g_0 = 9.8 \text{ m/s}^2$, $z_0 = 27 \text{ m}$
 $\lambda = 42^\circ$

$$\Rightarrow \boxed{y(t_{\text{FALL}}) \approx 2.29 \times 10^{-3} \text{ m}}$$



$$\vec{a}_{\text{tot}} = \vec{g}_0 + \vec{a}_{\text{cor}}$$

$$\vec{g}_0 = -g_0 \hat{z}$$

$$\vec{a}_{\text{cor}} = -2\vec{\omega} \times \vec{V}$$

Recall (from problem 2): $\vec{\omega} = -\omega \cos \lambda \hat{x} + \omega \sin \lambda \hat{z}$

generic velocity of projectile $\vec{V} = V_x \hat{x} + V_y \hat{y} + V_z \hat{z}$

Assuming "small deflections" from the vertical (z) due to Coriolis term $\Rightarrow V_y \approx 0$

Note that in this case $V_x \neq 0$ as the projectile has a significant V_x component (2d - projectile motion)

Note: Only make this assumption when calculating Coriolis term

$$\vec{\omega} \times \vec{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ -\omega \cos \lambda & 0 & \omega \sin \lambda \\ V_x & 0 & V_z \end{vmatrix} = -\hat{y}(-\omega \cos \lambda V_z - \omega \sin \lambda V_x)$$

$$\vec{a}_{\text{cor}} = a_{\text{cor}} \hat{y} = -2\omega (\cos \lambda V_z + \sin \lambda V_x) \hat{y}$$

$$\vec{a}_{\text{tot}} = -g_0 \hat{z} + a_{\text{cor}} \hat{y} \rightarrow \text{projectile motion is subject to two acceleration components to be treated separately!}$$

gravity $-\vec{g}_0 \hat{z}$ term:

$$\frac{dV_z}{dt} = -g_0 \Rightarrow \int_{V_{0,z}}^{V_z} dV_z = -g_0 \int_0^t dt \Rightarrow V_z = \underbrace{V_{0,z}}_{V_0 \sin \alpha} - g_0 t \Rightarrow \underline{V_z = V_0 \sin \alpha - g_0 t}$$

$$\int_0^t \frac{dz}{dt} dt = \int_0^t (V_0 \sin \alpha - g_0 t) dt \Rightarrow \underline{z(t) = V_0 \sin \alpha \cdot t - \frac{1}{2} g_0 t^2}$$

$$\Rightarrow \text{Calculate fall time: } z(t_{\text{FALL}}) = V_0 \sin \alpha t_{\text{FALL}} - \frac{1}{2} g_0 t_{\text{FALL}}^2 = 0$$

5) PART A continued...

Calculate fall time: $\frac{1}{2} g_0 t_{\text{FALL}}^2 = v_0 \sin \alpha t_{\text{FALL}}$

$$\Rightarrow t_{\text{FALL}} = \frac{2 v_0 \sin \alpha}{g_0}$$

Calculate "horizontal" range projectile travels:

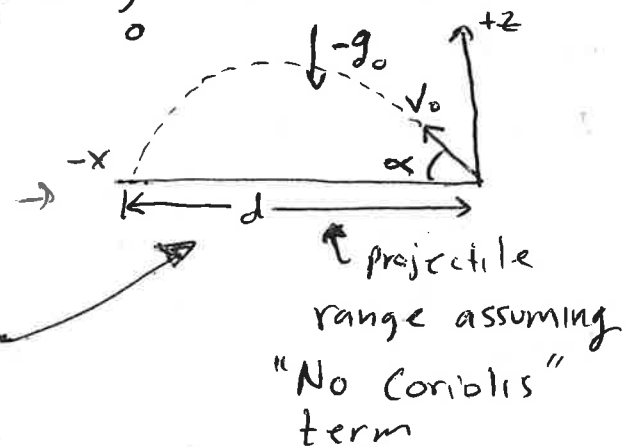
Note that since $a_x = 0$ (no x-component of acceleration)

$$\Rightarrow \frac{dv_x}{dt} = 0 \quad \underline{v_x = -v_0 \cos \alpha} \quad \int_0^t \frac{dx}{dt} dt = - \int_0^t v_0 \cos \alpha dt$$

$$\Rightarrow \underline{X(t) = -v_0 \cos \alpha \cdot t}$$

$$d \equiv X(t_{\text{FALL}}) = -v_0 \cos \alpha t_{\text{FALL}}$$

$$\Rightarrow d = \left| -\frac{2 v_0^2 \sin \alpha \cos \alpha}{g_0} \right|$$



Coriolis " $a_{\text{cor}} \hat{y}$ " term

From (x, z) components

$$a_{\text{cor}} = \frac{dv_y}{dt} = -2\omega (\cos \lambda v_z + \sin \lambda v_x) \Leftarrow \begin{cases} v_x = -v_0 \cos \alpha \\ v_z = v_0 \sin \alpha - g_0 t \end{cases}$$

$$= -2\omega (v_0 \sin \alpha \cos \lambda - g_0 \cos \lambda \cdot t - v_0 \cos \alpha \sin \lambda)$$

$$= -2\omega [v_0 (\sin \alpha \cos \lambda - \cos \alpha \sin \lambda) - g_0 \cos \lambda t] \quad \begin{array}{l} \text{Now use} \\ \sin \alpha = \cos \alpha = \frac{\sqrt{2}}{2} \\ \alpha = 45^\circ \text{ or } \frac{\pi}{4} \end{array}$$

$$a_{\text{cor}} = -\frac{\omega v_0 \sqrt{2}}{2} (\cos \lambda - \sin \lambda) + 2\omega g_0 \cos \lambda t$$

$$\int_0^t dv_y = -\frac{\omega v_0 \sqrt{2}}{2} (\cos \lambda - \sin \lambda) \int_0^t dt + 2\omega g_0 \cos \lambda \int_0^t t dt$$

5) PART A continued... Coriolis " $a_{cor} \hat{y}$ " term

$$\dot{v}_y = -\frac{\omega v_0 t \sqrt{2}}{2} (\cos \lambda - \sin \lambda) + \omega g_0 \cos \lambda t^2$$

$$\int_0^y dy = -\frac{\omega v_0 \sqrt{2}}{2} (\cos \lambda - \sin \lambda) \int_0^t t dt + \omega g_0 \cos \lambda \int_0^t t^2 dt$$

$\underbrace{\int_0^t t dt}_{t^2/2} \quad \underbrace{\int_0^t t^2 dt}_{t^3/3}$

$$y(t) = -\frac{\omega v_0 t^2 \sqrt{2}}{2} (\cos \lambda - \sin \lambda) + \frac{1}{3} \omega g_0 t^3 \cos \lambda$$

Calculate deflection due to "Coriolis effect"

Recall: $t_{\text{FALL}} = \frac{2v_0 \sin \alpha}{g_0} \Rightarrow \frac{2v_0 \sqrt{2}}{g_0 \cdot 2} \quad t_{\text{FALL}} = \frac{\sqrt{2} v_0}{g_0}$

$\alpha = \pi/4$
 $\sin \alpha = \frac{\sqrt{2}}{2}$

$$d = \left| -\frac{2v_0^2 \sin \alpha \cos \alpha}{g_0} \right| \Rightarrow \frac{-2v_0^2 \left(\frac{\sqrt{2}}{2}\right)^2}{g_0} \rightarrow \frac{-2v_0^2 \cdot \frac{2}{4}}{g_0} = \left| -\frac{v_0^2}{g_0} \right|$$

$\alpha = \pi/4$
 $\sin \alpha = \frac{\sqrt{2}}{2}$
 $\cos \alpha = \frac{\sqrt{2}}{2}$

$$t_{\text{FALL}} = \frac{\sqrt{2} v_0}{g_0}$$

$$d = \frac{v_0^2}{g_0} \rightarrow v_0 = \sqrt{d g_0} \Rightarrow t_{\text{FALL}} = \frac{\sqrt{2} \sqrt{d g_0}}{g_0} = \frac{\sqrt{2d} \sqrt{g_0}}{g_0} \cdot \frac{\sqrt{g_0}}{\sqrt{g_0}}$$

$$\Rightarrow \boxed{t_{\text{FALL}} = \sqrt{\frac{2d}{g_0}}}$$

$\Rightarrow$ Substitute t_{FALL} in $y(t=t_{\text{FALL}})$ to calculate deflection.

5) PART A continued...

$$Y(t_{\text{FALL}}) = -\frac{\omega v_0 \sqrt{2d}}{2\sqrt{2} g_0} (\cos \lambda - \sin \lambda) + \frac{1}{3} \omega \cancel{g_0} (\cos \lambda) \frac{2d}{g_0} \left(\frac{2d}{g_0}\right)^{1/2}$$

Recall $\rightarrow v_0 = \sqrt{g_0 d}$

$$= -\frac{\omega \sqrt{g_0 d} \cdot 2d}{2\sqrt{2} \sqrt{g_0^2}} (\cos \lambda - \sin \lambda) + \frac{2}{3} \omega d \sqrt{\frac{2d}{g_0}} \cos \lambda$$

$$= -\omega d \sqrt{\frac{2d}{g_0}} (\cos \lambda - \sin \lambda) + \frac{2}{3} \omega d \sqrt{\frac{2d}{g_0}} \cos \lambda$$

$$\Rightarrow \frac{2}{3} \omega d \sqrt{\frac{2d}{g_0}} \cos \lambda - \omega d \sqrt{\frac{2d}{g_0}} \cos \lambda + \omega d \sqrt{\frac{2d}{g_0}} \sin \lambda$$

$$\Rightarrow Y(t_{\text{FALL}}) = \omega d \sqrt{\frac{2d}{g_0}} \left(\underbrace{\frac{2}{3} \cos \lambda - \frac{3}{3} \cos \lambda}_{-\frac{1}{3} \cos \lambda} + \sin \lambda \right)$$

$$\Rightarrow Y(\lambda) = \omega d \sqrt{\frac{2d}{g_0}} \left(\sin \lambda - \frac{1}{3} \cos \lambda \right)$$

where $t_{\text{FALL}} \equiv \sqrt{\frac{2d}{g_0}}$, $d \equiv v_0^2 / g_0$

PART B: $\lambda = 30^\circ$, $d = 15000 \text{ m}$, $\omega \approx 7.3 \times 10^{-5} \text{ rad/s}$

$$Y(\lambda = 30^\circ) \approx 12.8 \text{ m}$$